

# Scalability & Future Plan

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|----------------------|--------------------------------------------|
| <b>Date</b>          | 15 March 2026                              |
| <b>Team ID</b>       | XXXXXX                                     |
| <b>Project Name</b>  | Voice-Based Concept Understanding Analyzer |
| <b>Maximum Marks</b> | 1 Mark                                     |

## Scalability & Future Plan

This document explains how the Voice-Based Concept Understanding Analyzer can be expanded to support a larger number of users, improved performance, cloud deployment, enhanced security, and additional Artificial Intelligence features in future releases.

## Current System Limitations

| S.No | Limitation                           | Impact                                         | Priority |
|------|--------------------------------------|------------------------------------------------|----------|
| 1    | Supports only single-user execution. | Cannot handle multiple concurrent users.       | High     |
| 2    | Uses local JSON storage.             | Limited scalability and centralized storage.   | High     |
| 3    | Limited AI evaluation features.      | Cannot evaluate advanced communication skills. | Medium   |

## Scalability Plan

| S.No | Scalability Aspect | Current State                      | Proposed Upgrade / Solution                                       |
|------|--------------------|------------------------------------|-------------------------------------------------------------------|
| 1    | User Load          | Single-user Streamlit application. | Deploy on cloud with multi-user support and load balancing.       |
| 2    | Data Storage       | JSON files.                        | Use PostgreSQL or Firebase for centralized storage.               |
| 3    | Performance        | CPU-based processing.              | Optimize AI inference and enable GPU acceleration.                |
| 4    | Security           | Basic local authentication.        | Add user authentication, encrypted storage and role-based access. |

## Future Roadmap

| Phase   | Planned Feature / Enhancement                    | Target Timeline | Expected Impact                                     |
|---------|--------------------------------------------------|-----------------|-----------------------------------------------------|
| Phase 1 | Cloud Deployment using Streamlit Community Cloud | 1 Month         | Online accessibility for users.                     |
| Phase 2 | Database Integration (PostgreSQL/Firebase)       | 2 Months        | Reliable and scalable data management.              |
| Phase 3 | Advanced AI-based Speech & Emotion Analysis      | 4 Months        | Improved evaluation accuracy and detailed feedback. |
| Phase 4 | Mobile Application & Multi-language Support      | 6 Months        | Increased accessibility and wider user adoption.    |